Reflected Blocks, Implicit Parameters, and Full Geometric Recovery

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Abstract

This supplement gives a complete proof of the reflected-block upper construction and of the implicit parameters defining $U_4 = C_4(1/20000)$. It includes the all-pairs angular criterion, recovery by actual integer labels, a quantitative radius estimate, uniqueness of the successive parameters, and the rational gates used to isolate them. The universal statements are analytic theorems; the accompanying bounded rational computations establish specified constant inequalities. The construction is an upper bound and does not assert an optimal partition or a finite geometric optimum. These are existing results reorganized from the source proofs listed in the dependency document.

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1 The Geometric Problem and the Target Bound

Circles of radii $1, \dots, n$ are externally tangent to a central circle of radius $R > 0$. Let $R^*(n)$ be the infimum of feasible central radii. For a prescribed cyclic order σ, let $R_{\mathrm{full}}(\sigma)$ use all pairs and let

$R_{\mathrm{chain}}(\sigma)$ use only adjacent gaps. The cosine law gives the required angle

$$\theta_R(a,b) = 2\arcsin\sqrt{\frac{ab}{(R+a)(R+b)}}. \tag{1}$$

Both directed arcs between each pair must be at least this angle. This is equivalent to Cartesian non-overlap, since their sum is 2π and the shorter angle lies in $[0,\pi]$. Throughout, inverse trigonometric functions have their principal real branches. The kernel increases in each outer radius and decreases strictly in R. In general $R_{\mathrm{chain}} \le R_{\mathrm{full}}$; no equality is presumed here.

Define, on $\mathcal{D} = [0,1]\times[1,2]^2$,

$$g(t,X,Y) = \max\{\sqrt{t}(\sqrt{X}+\sqrt{Y}), \sqrt{XY}\}, \qquad C(\mu) = \frac{1}{4\pi}\int g\,\mathrm{d}\mu. \tag{2}$$

Sections 4–6 uniquely define $x_*, \widehat{\alpha}, \epsilon_b, \Delta_*$. Set $A = 1+\widehat{\alpha}$, $\lambda = Ax_*$, $v = \lambda + \epsilon_b$, and $w = v + \Delta_*$. The measure $\mu_4(\eta)$ consists of four separate symmetric reflections on the consecutive intervals of lengths $(\lambda, \epsilon_b, \Delta_*, \eta)$, followed by the shifted diagonal; its precise general definition is (6). Put $C_3(d) = C(\mu_{\widehat{\alpha};(\lambda,\epsilon_b,d)})$ and $C_4(\eta) = C(\mu_4(\eta))$. The final result proved here is

$$\limsup_{n\to\infty}\frac{R^*(n)}{n^2} \le U_4 := C_4(1/20000) < C_3(\Delta_*) - \frac{1}{4608000000000000}. \tag{3}$$

If C_* denotes the global limit established in the main article, the left side is C_*. Its existence is not needed for this supplement's upper-bound proof. Terminating decimal bounds below are exact rationals, never definitions of the implicit parameters.

2 An Exact All-Pairs Criterion

Lemma 2.1 (High-Shell Inequality)**.** *If $0 < u \le v \le 2u$, then $2\theta_R(u,u) > \theta_R(v,v)$. Consequently every path $H_0,\dots,H_j$ in $[u,v]$, $j \ge 1$, satisfies $\sum_{i=1}^{j}\theta_R(H_{i-1},H_i) \ge \theta_R(H_0,H_j)$.*

Proof. Write $x = u/(R+u)$ and $z = v/(R+v) \le 2x/(1+x)$. If $x \ge 1/\sqrt{2}$, then $2\arcsin x \ge \pi/2 > \arcsin z$. Otherwise

$$(1+x)^2(1-x^2) - 1 = x(2-2x^2-x^3) > 0,$$

because $x^2 < 1/2$, $x < 3/4$, and $2-2x^2-x^3 > 5/8$. Thus $z < 2x\sqrt{1-x^2} = \sin(2\arcsin x)$. Both comparison angles are in $(0,\pi/2)$, so the same strict inequality follows. Multiplying by two proves the first assertion. For any $a,b,c \in [u,v]$,

$$\theta_R(a,b) + \theta_R(b,c) \ge 2\theta_R(u,u) > \theta_R(v,v) \ge \theta_R(a,c).$$

Induction contracts a path. The one-edge case is equality. □

Let $m \ge 2$ and let $(P_1,\dots,P_m)$ be any permutation of $\{m+1,\dots,2m\}$. Subscripts are cyclic, with $P_0 = P_m$. The order is $(1,P_1,2,P_2,\dots,m,P_m)$. Define

$$a_i = \theta_R(P_{i-1},i), \quad b_i = \theta_R(i,P_i), \quad c_i = \theta_R(P_{i-1},P_i), \quad d_i = \max\{a_i+b_i, c_i\}. \tag{4}$$

Theorem 2.2 (Full Cell Criterion)**.** *The order is feasible exactly when $S_P(R) := \sum_i d_i \le 2\pi$. More precisely its feasible gap vectors are exactly*

$$x_i \ge a_i, \quad y_i \ge b_i, \quad x_i + y_i \ge c_i, \qquad \sum_i (x_i+y_i) = 2\pi, \tag{5}$$

where x_i is the gap $P_{i-1}\to i$ and y_i is $i \to P_i$. The full minimum is the unique positive root ρ_P of $S_P(R) = 2\pi$.

Proof. Necessity follows from the two adjacent constraints and the high-pair constraint on each whole cell. For sufficiency first take any positive gaps satisfying the three local inequalities, without requiring closure. For each pair choose either simple cyclic path; the following arguments apply separately in both directions.

For two highs, the path consists of whole cells. Replacing their lengths by c_i and applying Lemma 2.1 on $[m+1, 2m]$ gives the pair bound. For a low i and a high P_j, the first high on the path is a neighbor U. If $U = P_j$, use adjacency. Otherwise the remaining whole-cell high path has length at least $\theta_R(U, P_j) > \theta_R(i, P_j)$. For two distinct lows i, j, let U and V be the first and last highs. If $U = V$, the simple path has two edges and its first gap alone is at least $\theta_R(i, U) > \theta_R(i, j)$. If $U \neq V$, the middle high path is at least $\theta_R(U, V) > \theta_R(i, j)$. These cases exhaust all pairs and both arcs, including paths crossing the cyclic cell $P_m \to 1 \to P_1$. For $m = 2$ each directed high path is one whole cell; no third high is required. The low-high and low-low arguments remain valid.

When $S_P(R) \le 2\pi$, take $x_i = a_i$, $y_i = d_i - a_i$ and add the remaining $2\pi - S_P(R)$ to any single gap. Each directed path either increases or stays fixed; both bounds survive and closure holds. This also proves sufficiency of every vector in (5). Each of $a_i + b_i, c_i$ is positive, continuous and strictly decreasing in R; their maximum is strictly decreasing by choosing an attaining branch at the larger radius. Moreover $S_P(0+) = 2\pi m$ and $S_P(\infty) = 0$. The intermediate value theorem and monotonicity give the unique root and its feasible placement. □

3 Recovery of General Reflected Blocks

Fix $0 \le \alpha < 1$ and nonnegative lengths $\ell_1, \dots, \ell_k$, $k \ge 1$, with $T = \sum_j \ell_j < 1 - \alpha$. Put $t_0 = 0$, $t_j = \sum_{h \le j} \ell_h$, $A = 1 + \alpha$, and $h_\alpha(t) = 1 + \{t + \alpha\}$. Define a probability measure by

$$\begin{aligned} \int F \, \mathrm{d}\mu_{\alpha;\ell} = \frac{1}{2} \sum_{j=1}^{k} \int_{t_{j-1}}^{t_j} \big[& F(t, A + t, A + t_{j-1} + t_j - t) \\ & + F(t, A + t_{j-1} + t_j - t, A + t) \big] \, \mathrm{d}t \\ + \int_T^1 & F(t, h_\alpha(t), h_\alpha(t)) \, \mathrm{d}t. \end{aligned} \tag{6}$$

Each high marginal and the low marginal are uniform on their respective intervals. Each slab reflection is a measure-preserving involution, and the unchanged shift is a bijection modulo its wrap. Zero lengths and endpoint assignments do not affect the measure. Adjacent slabs are reflected separately, never as their union.

Theorem 3.1 (Quantitative Full Transfer)**.** *Let $I = \int g \, \mathrm{d}\mu_{\alpha;\ell}$ and $M_0 = \max\{2, \lceil 2/(1 - \alpha - T) \rceil\}$. There are genuine orders on $1, \dots, 2m$ with full minimum ρ_m for every $m \ge M_0$, and, for $m \ge \max\{M_0, 2048\}$,*

$$\left| \frac{\rho_m}{(2m)^2} - \frac{I}{4\pi} \right| \le \frac{1}{4\pi} \left(\frac{6k^2 + 28k + 1060}{m} + \frac{16384}{3m^2} \right). \tag{7}$$

Consequently $\limsup_n R^*(n)/n^2 \le C(\mu_{\alpha;\ell})$. No chord-dominance assumption on any slab is required.

Proof. Define exact integers

$$s = \lfloor \alpha m \rfloor, \quad l_j = 2\lfloor \ell_j m / 2 \rfloor, \quad a_0 = 0, \quad a_j = \sum_{h \le j} l_h, \quad z = a_k, \quad r = m - s,$$

and $H(u) = m + 1 + ((u + s - 1) \bmod m)$. On the block $(a_{j-1}, a_j]$ put $J(i) = 2a_{j-1} + l_j + 2 - i$ when i is even, and $J(i) = i$ otherwise. Set $P_i = H(J(i))$, $P_0 = P_m$. Each block reverses its even ranks and fixes its odd ranks, giving a disjoint involution; H is a bijection. Lengths zero and two cause no exception. Since $s + z \le m(\alpha + T)$, $r - z \ge 2$. The blocks and their exits precede the high wrap.

For completeness the cells have the following exhaustive inventory. A nonempty block $(a, a + l]$ has interior indices $a + 2, \dots, a + l$. The predecessor/current pair there is

$$\begin{cases} (m + s + i - 1, m + s + 2a + l + 2 - i), & i \text{ even}, \\ (m + s + 2a + l + 3 - i, m + s + i), & i \text{ odd}. \end{cases}$$

The exceptional set is $\{1, r, r + 1\} \cup \{a_j + 1 : l_j > 0\}$, intersected with $\{1, \dots, m\}$, with duplicates counted once. It has at most $k + 3$ members. Cell 1 is $(H(m), H(1))$; the exit from $(a, a + l]$ is $(H(a + 2), H(a + l + 1))$. It is also the next nonempty block's entry even if zero lengths intervene. Cell r is $(2m - 1, 2m)$, and cell $r + 1$, when present, is $(2m, m + 1)$. All other cells are ordinary $(H(i - 1), H(i))$. The interiors exclude every shared entry index, and the wrap lies outside them, proving disjointness and completeness.

Let $\mu_m = m^{-1} \sum_i \delta_{(i/m, P_{i-1}/m, P_i/m)}$. For a continuous, possibly nonsymmetric F, write $B_F = \|F\|_\infty$ and ω_F for its max-norm modulus. Replace α, ℓ_j in (6) by $s/m, l_j/m$ to obtain $\bar{\mu}_m$. On each double panel $[(a + 2h - 2)/m, (a + 2h)/m]$, assign half the first-orientation integral to its even cell and half the reverse-orientation integral to its odd cell. Each mass is $1/m$. Interior coordinate errors are at most $4/m$; ordinary diagonal panels have errors at most $1/m$. Exceptional panel assignments each cost at most $2B_F/m$, so

$$\left| \int F \, \mathrm{d}(\mu_m - \bar{\mu}_m) \right| \le \omega_F(4/m) + 2(k + 3)B_F/m.$$

The floor errors are less than $1/m$ for the shift and $2/m$ for each length. The intervals between a_j/m and t_j, together with the wrap interval, have total length at most $[k(k + 1) + 1]/m$. This bound allows overlap and zero slabs. Off these intervals the same slab and orientation apply; the reflected coordinate error in slab j is less than $(4j - 1)/m$ and the other is less than $1/m$. Thus

$$\left| \int F \, \mathrm{d}(\mu_m - \mu) \right| \le \omega_F(4/m) + \omega_F((4k - 1)/m) + \frac{(2k^2 + 4k + 8)B_F}{m}. \tag{8}$$

The complete maximum g obeys $1 \le g < 3$ and is 4-Lipschitz. For $t \le 1/4$ the chord dominates because $\sqrt{t/X} + \sqrt{t/Y} \le 2\sqrt{t} \le 1$. Above $1/4$, the sum of absolute partial derivatives of the chain is at most $2\sqrt{2} + 1 < 4$, and that of the chord is less than two. The maximum of these Lipschitz functions has the same bound; splitting at $t = 1/4$ proves it globally. Putting $G_m = \int g \, \mathrm{d}\mu_m$, (8) yields

$$|G_m - I| \le (6k^2 + 28k + 36)/m. \tag{9}$$

This estimate retains every finite maximum and seam even if its branch differs from its continuum counterpart.

By Theorem 2.2, the actual full minimum is the root of $S_m(R) = \sum_i \max\{\theta_R(P_{i-1}, i) + \theta_R(i, P_i), \theta_R(P_{i-1}, P_i)\} = 2\pi$. For $R = 4cm^2$, $c \ge 1/32$, $m \ge 32$, and $1 \le u, v \le 2m$, put $q = \sqrt{uv}/R$. Then $q \le 1/(2cm) \le 1/2$. The relative denominator change satisfies

$$0 \le 1 - \frac{1}{\sqrt{(1 + u/R)(1 + v/R)}} \le \frac{u + v}{2R}.$$

Also $0 \le \arcsin t - t \le t^3/3$ for $0 \le t \le 1/2$, by integrating $1/\sqrt{1-t^2} - 1 \le t^2$ on that interval. It follows that

$$\left|\theta_R(u,v) - \frac{2\sqrt{uv}}{R}\right| \le \frac{512}{m^2} + \frac{8192}{3m^3}, \qquad \left|S_m(4cm^2) - \frac{G_m}{2c}\right| \le E_m := \frac{1024}{m} + \frac{16384}{3m^2}.$$

The last step uses the one-Lipschitz property of the scalar maximum and two angles per chain. For $m \ge 2048$, $E_m < 1$; at $c = 1/32$ the score exceeds $15 > 2\pi$, while at $c = 1/2$ it is below $4 < 2\pi$. Thus the actual normalized root $c_m = \rho_m/(2m)^2$ is first bracketed in $(1/32, 1/2)$. Substitution then gives $|4\pi c_m - G_m| \le 2c_m E_m \le E_m$. Together with (9) this is (7). The root placement is all-pairs feasible. Deleting only label $2m$ keeps exactly $1, \dots, 2m-1$ and all their constraints; the normalization factor $(2m/(2m-1))^2$ tends to one. This proves the global limsup bound for both parities without identifying an odd fixed-order minimum. □

The same construction allows countably many lengths with total $T < 1 - \alpha$. Truncation after k blocks and reversion of the tail to the diagonal changes any F integral by at most $2B_F \sum_{j>k} \ell_j$. For g the bound is six times that tail length. Apply the finite theorem and then let $k \to \infty$. A single recovery sequence also results by choosing increasing thresholds M_k growing faster than k^3, each satisfying the common geometric gate, and using the largest $k \le m$ with $M_k \le m$. Both (8) for Lipschitz tests and the omitted tail vanish; density of Lipschitz functions on compact $\mathcal{D}$ gives continuous-test recovery. This supplies countable upper constructions, not an optimum over partitions.

4 The Exact Normalized Prefix Parameter

For $0 \le u \le x \le 1$, define

$$c(u,x) = \sqrt{(1+u)(1+x-u)},$$
$$k(u,x) = \sqrt{u}(\sqrt{1+u} + \sqrt{1+x-u}),$$
$$d(u) = \max\{1+u, 2\sqrt{u(1+u)}\}.$$

$$E(x) = \int_0^x [\max\{c(u,x), k(u,x)\} - d(u)]\,\mathrm{d}u. \tag{10}$$

Theorem 4.1. *The function E has a unique global minimum x_* on $[0,1]$, and $719/2500 < x_* < 2877/10000$. These are the exact bounds $0.2876 < x_* < 0.2877$.*

Proof. The ratio $v(u,x) = k/c = \sqrt{u/(1+u)} + \sqrt{u/(1+x-u)}$ strictly increases in u. Its endpoint equals one at the unique $\tau \in (1/4, 1/3)$ solving $\sqrt{\tau/(1+\tau)} + \sqrt{\tau} = 1$. Set $z(x) = x$ for $x \le \tau$; for $x > \tau$ let $z \in (0,x)$ be the unique unsquared root $v(z,x) = 1$. The block is chord up to z, then chain. The diagonal switches at $u = 1/3$. For $x > \tau$, with $y = 1 + x - z$ and $r = \sqrt{z/(1+z)}$,

$$z' = \frac{z}{(y/(1+z))^{3/2} + 1 + x} = \frac{(1-r)^2}{2 - r + 2r^2} > 0. \tag{11}$$

The second equality uses $y = z/(1-r)^2$ and $z = r^2/(1-r^2)$. All denominators are positive.

Let $\Phi = E'$. Cost equality at the switch cancels its first moving boundary term. The endpoint ties at $x = \tau, 1/3$ also agree, so E is C^1 there. On the chord interval,

$$\Phi(x) = \sqrt{1+x} + (1+x/2)\arcsin\frac{x}{2+x} - (1+x),$$

$$\Phi(0) = \Phi'(0) = 0, \qquad \Phi''(x) = -\frac{1}{2(2+x)(1+x)^{3/2}} < 0.$$

Thus $\Phi < 0$ on $(0, \tau]$. On $x > \tau$, put

$$I_1 = \int_0^z \sqrt{\frac{1+u}{1+x-u}}\,\mathrm{d}u, \qquad I_2 = \int_z^x \sqrt{\frac{u}{1+x-u}}\,\mathrm{d}u, \qquad p = \sqrt{x}(\sqrt{1+x}+1).$$

Differentiation yields

$$\Phi = p - d(x) + (I_1 + I_2)/2, \qquad \Phi' = p' - d'(x) + \sqrt{x}/2 - J/4 + Q, \tag{12}$$

where

$$J = \int_0^z \frac{\sqrt{1+u}}{(1+x-u)^{3/2}}\,\mathrm{d}u + \int_z^x \frac{\sqrt{u}}{(1+x-u)^{3/2}}\,\mathrm{d}u, \quad Q = \frac{z'(\sqrt{1+z}-\sqrt{z})}{2\sqrt{1+x-z}} > 0.$$

For $\tau < x < 1/3$, $d' = 1$, $J \le x\sqrt{1+x} < 7/18$ and $p' > 1/\sqrt{x} \ge \sqrt{3} > 5/3$. Hence $\Phi' > 41/72 > 0$. At $x = 1/3$, $v(2/7, x) < 1$: the pre-square residual is $50/99 > 0$ and its square minus $4(2/9)(3/11)$ is $124/9801 > 0$. Thus $z > 2/7$, and

$$\Phi(1/3) > \frac{1}{\sqrt{3}} - \frac{2}{3} + \frac{\sqrt{3}}{14} = \frac{17\sqrt{3}}{42} - \frac{2}{3} > 0$$

by $17^2 \cdot 3 > 28^2$. There is exactly one zero x_* in $(\tau, 1/3)$, with strict decrease then increase of E on $[0, 1/3]$.

Global minimality needs a separate tail argument; convexity must not be extrapolated across $1/3$. On $1/3 < x \le 1$, (12) becomes

$$\Phi' = -M - J/4 + Q, \qquad M = \frac{1 + 2x - (1+x)^{3/2}}{2\sqrt{x(1+x)}}.$$

Since $z > 2/7$, $r > \sqrt{2/9} > 8/17$. The polynomial $5 - 13r + 5r^2$ decreases on $(0, 1)$ and is $-3/289$ at $8/17$; (11) gives $z' < 1/7$. Also $\sqrt{1+z} - \sqrt{z} < 2/3$, so $Q < 1/21$. Write $q = \sqrt{1+x} \in [2/\sqrt{3}, \sqrt{2}]$. Then

$$M = \sqrt{\frac{q-1}{q+1}}\frac{1+q-q^2}{2q}.$$

For $q \le 4/3$ the factors exceed $1/4$ and are at least $5/24$, respectively, so $M > 5/96 > 1/20$. For $q \ge 4/3$ they are at least $1/\sqrt{7} > 3/8$ and $(2-\sqrt{2})/4 > 1/7$, so again $M > 1/20$. These comparisons follow by positive squares, using $2/\sqrt{3} > 17/15$ and $\sqrt{2} < 10/7$. Therefore $\Phi' < -1/420$ throughout the tail. The rational gates in Section 7 give $E(1) > 3/250 > 0$. Strict concavity makes E at least its endpoint chord on $[1/3, 1]$. Both endpoint values exceed $E(x_*) < 0$, proving unique global minimality. The same gates give opposite signs at $719/2500$ and $2877/10000$, both above τ, locating this unique root. They also give $\Phi(4/5) < 0$; thus a later local maximum exists, and does not invalidate the endpoint-chord argument. □

5 The Exact Shift Parameter

For $0 \le \alpha \le 1/2$ let $A = 1+\alpha$, $a = A/3$, $b = 1-\alpha$, and

$$K(\alpha) = \int_0^1 \max\{\sqrt{t h_\alpha(t)}, h_\alpha(t)/2\}\,\mathrm{d}t, \qquad e = E(x_*), \qquad F(\alpha) = K(\alpha) + A^2 e/2.$$

The full prefix cost at length Ax_* is $F(\alpha)/(2\pi)$. Indeed $Ax_* < a \le b$, the replaced diagonal is chord, and scaling $t = Au$ gives its increment $A^2E(x_*)$. Moreover $b - Ax_* > 1/2 - (3/2)(2877/10000) = 1369/20000 > 1/15$, so the reflection stays before the wrap even at the closed endpoints.

Theorem 5.1. *There is exactly one zero $\widehat{\alpha}$ of $F'(\alpha) = K'(\alpha) + (1+\alpha)e$ on $[0, 1/2]$. It is the unique minimum of this fixed-x_* prefix family, and $1093/10000 < \widehat{\alpha} < 10931/100000$.*

Proof. The split integral is

$$K = \int_0^a (A+t)/2\,\mathrm{d}t + \int_a^b \sqrt{t(A+t)}\,\mathrm{d}t + \int_b^1 \sqrt{t(\alpha+t)}\,\mathrm{d}t.$$

Differentiation, including the moving wrap, gives

$$D(\alpha) := K'(\alpha) = \frac{a}{2} + \frac{1}{2}\int_a^b \sqrt{\frac{t}{t+A}}\,\mathrm{d}t + \frac{1}{2}\int_b^1 \sqrt{\frac{t}{t+\alpha}}\,\mathrm{d}t - (\sqrt{2}-1)\sqrt{b}. \tag{13}$$

For $0 < x < 1/3$, centering the chord at $M = 1 + x/2$ gives $E(x) \ge -\int_{-x/2}^{x/2} u^2/M\,\mathrm{d}u = -x^3/(12M) > -1/324$. At $x = 1/4$, which is strictly chord, rationalization gives $-E(1/4) > 1/1728$. Theorem 4.1 implies $-1/324 < e < -1/1728$. A second differentiation of (13) gives

$$K'' = \frac{1}{12} + \frac{(1-1/\sqrt{2})\sqrt{b}}{2} + \frac{\sqrt{2}-1}{2\sqrt{b}} - \frac{T_1}{4}, \quad T_1 = \int_a^b \frac{\sqrt{t}}{(t+A)^{3/2}}\,\mathrm{d}t + \int_b^1 \frac{\sqrt{t}}{(t+\alpha)^{3/2}}\,\mathrm{d}t.$$

Both denominator bases are at least one; $T_1 \le 1 - a \le 2/3$. Using $1/2 \le b \le 1$ and $\sqrt{2} > 7/5$ yields $K'' > 7/60$, so $F'' > 7/60 - 1/324 = 46/405 > 1/9$ on the whole domain.

At zero, $D(0) = 5/6 - \sqrt{2}/2 - \frac{1}{2}\log((1+\sqrt{2})/\sqrt{3})$. For $z > 1$, $\log z > 2(z-1)/(z+1)$ by differentiation. Since the argument exceeds $48/35$, using $\sqrt{3} < 7/4$ gives $F'(0) < 5/6 - 7/10 - 13/83 = -29/1245 < 0$. At $1/2$ the surviving integrand in (13) is at least $1/\sqrt{2}$, so $D(1/2) \ge 5/(4\sqrt{2}) - 3/4 > 1/8$ and $F'(1/2) > 1/8 - (3/2)/324 = 13/108 > 0$. Strict convexity proves uniqueness and minimality.

For the tighter location set $x_0 = 5753/20000$, the midpoint of the x_* bracket. On $(\tau, 1/3)$, (12) gives $E'' < 8/3 - 1 + 1/3 + 1/8 = 17/8 < 3$: here $p' < 8/3$, $z' < x/(1+x) < 1/4$, $Q < 1/8$, and $J \ge 0$. Taylor's theorem at the stationary minimum gives

$$E(x_0) - 3/800000000 < e \le E(x_0). \tag{14}$$

The directed gates in Section 7, evaluating only this rational argument and the two rational α endpoints, give $F'(1093/10000) < 0 < F'(10931/100000)$. These isolate the same unique zero. □

6 A Common Width Lemma and the Next Two Parameters

Fix $A > 1$ and $A/4 < u < A/3$. Put $a = A/3$, $B = A + u$, and $h = a - u$, so $B > 5A/4$, $0 < h < A/12$, $h/B < 1/15$. Reflect a diagonal slab $[u, u + d]$, $0 \le d \le h$, separately from all preceding slabs. Its unnormalized complete-cost increment is exactly

$$D_{A,u}(d) = \int_0^d \big[\max\{\sqrt{(B+s)(B+d-s)}, \sqrt{u+s}(\sqrt{B+s} + \sqrt{B+d-s})\} - (B+s)\big]\,\mathrm{d}s. \tag{15}$$

The removed diagonal is chord because $A - 3(u+s) \ge 0$; its only possible tie is $s = d = h$. Reflection preserves each marginal as in (6), and unchanged slabs cancel with their full maxima.

Lemma 6.1 (Unique Mixed Width)**.** *The function $D_{A,u}$ has a unique minimum d_* on $[0, h]$. Define τ_u as the unique unsquared root in $(0, h)$ of*

$$W(d) = \sqrt{\frac{u+d}{B+d}} + \sqrt{\frac{u+d}{B}} = 1.$$

Then $\tau_u < d_* < \tau_u + \tau_u^2/(8B)$, and d_* is the unique mixed-branch stationary point. The cost decreases strictly up to d_* and increases strictly afterwards.

Proof. Set $t = u + s$, $X = B + s$, $Y = B + d - s$, $c = \sqrt{XY}$, $k = \sqrt{t}(\sqrt{X} + \sqrt{Y})$, and $V = k/c$. Then

$$V_s = \frac{A/X^{3/2} + (Y+t)/Y^{3/2}}{2\sqrt{t}} > 0.$$

For $d > 0$, $V(0, d) < 1$ and $V(d/2, d) = 2\sqrt{(u + d/2)/(A + u + d/2)} < 1$. The endpoint W strictly increases, with $W(0) < 1$ and $W(h) = 1/2 + \sqrt{a/B} > 1$. This proves the asserted entry width. For $d \le \tau_u$ set $z = d$. For $d > \tau_u$ there is a unique $z \in (d/2, d)$ with $V(z, d) = 1$. The cost is

$$D_{A,u} = \int_0^z c\,\mathrm{d}s + \int_z^d k\,\mathrm{d}s - Bd - d^2/2.$$

Let $\Psi = D'_{A,u}$. In the all-chord regime,

$$\Psi(d) = B\phi(d/B), \qquad \phi(q) = \sqrt{1+q} + (1 + q/2)\arcsin\frac{q}{2+q} - (1+q).$$

Here $\phi(0) = \phi'(0) = 0$ and $-1/4 < \phi''(q) = -[2(2+q)(1+q)^{3/2}]^{-1} < 0$ for $q > 0$. Therefore $-\tau_u^2/(8B) < \Psi(\tau_u) < 0$ and the positive-width chord cost is strictly decreasing.

In the mixed regime put $q = u + d$, $p = \sqrt{q}(\sqrt{B+d} + \sqrt{B})$, $I_c = \int_0^z \sqrt{X/Y}\,\mathrm{d}s$ and $I_k = \int_z^d \sqrt{t/Y}\,\mathrm{d}s$. Leibniz differentiation gives

$$\Psi = p - (B+d) + (I_c + I_k)/2, \qquad z' = \frac{t}{A(Y/X)^{3/2} + Y + t} \in (0, 1/2), \tag{16}$$

where t, X, Y in z' are evaluated at $s = z$. The first moving-boundary term cancels by $c(z) = k(z)$. The bound on z' follows from $Y - t = A + d - 2z > A - d > 0$. The cost and first derivative are continuous at entry: the chain interval shrinks to zero and the endpoint costs agree. Equivalently put $s = dr$ on $[0, 1]$; positive radicands bound the difference quotients and the tie set has measure zero, permitting dominated differentiation. This also justifies the one-sided derivative at h.

The second derivative retains a nonzero moving-switch term:

$$\Psi' = p' - 1 + \tfrac{1}{2}\sqrt{q/B} - J/4 + Q, \quad p' = \frac{A+2q}{2\sqrt{q(A+q)}} + \frac{\sqrt{B}}{2\sqrt{q}}, \tag{17}$$

where

$$J = \int_0^z \frac{\sqrt{X}}{Y^{3/2}}\,\mathrm{d}s + \int_z^d \frac{\sqrt{t}}{Y^{3/2}}\,\mathrm{d}s, \qquad Q = \frac{z'(\sqrt{B+z} - \sqrt{u+z})}{2\sqrt{B+d-z}} > 0.$$

The first term in p' decreases in q and equals $5/4$ at $q = a$. Thus $p' > 5/4 + \sqrt{15}/4 > 17/8$. Also $0 < J \le d\sqrt{B+d}/B^{3/2} < 2/15$, giving $\Psi' > 131/120 > 1$ everywhere in the mixed regime. Cost equality cannot be used to drop Q, since the derivative integrands at the switch do not coincide.

At h, $B = 4a - h$, $u = a - h > 3a/4$, and

$$\Psi(h) = \sqrt{aB} - 2a + (I_c + I_k)/2, \qquad 2a - \sqrt{aB} = \frac{h}{2 + \sqrt{B/a}} < 2h/7.$$

On this slab $15a/4 < X, Y \le 4a$ and $3a/4 < t \le a$, so $\sqrt{X/Y} > 9/10$ and $\sqrt{t/Y} > 2/5$. Since $z(h) > h/2$, $(I_c + I_k)/2 > 13h/40$, whence $\Psi(h) > 11h/280 > 0$. Continuity and strict increase of Ψ in the mixed regime give its unique zero d_*. The chord sign excludes earlier stationary points, proving the unique global minimum including both endpoints. Finally integrating $\Psi' > 1$ from τ_u to d_* gives $d_* - \tau_u < -\Psi(\tau_u) < \tau_u^2/(8B)$. □

We now fix $A = 1 + \widehat{\alpha}$ and $\lambda = Ax_*$. Define

$$\epsilon_b = \operatorname*{argmin}_{0 \le d \le A/3 - \lambda} D_{A,\lambda}(d). \tag{18}$$

Lemma 6.1 proves this is a unique mixed stationary minimum. The rational gates below give

$$43/1000 < \tau_\lambda < 87/2000, \quad 43/1000 < \epsilon_b < 87/2000 + (87/2000)^2/8 < 7/160.$$

In particular $v = \lambda + \epsilon_b$ remains in $(A/4, A/3)$. Define in turn

$$\Delta_* = \operatorname*{argmin}_{0 \le d \le A/3 - v} D_{A,v}(d). \tag{19}$$

The same lemma proves its uniqueness and identifies its stationarity with $4\pi C_3'(\Delta_*) = 0$. The second pair of rational gates gives

$$\begin{gathered} 29/5000 < \tau_v < 67/10000, \\ 29/5000 < \Delta_* < 67/10000 + (67/10000)^2/8 < 27/4000 < A/3 - v. \end{gathered} \tag{20}$$

These definitions retain all mixed branches. They do not use a chord-only stationary equation or a rounded parameter.

7 Rational Gates and Their Reproduction

This section supplies the finite arithmetic premises used above. Every checker in the accompanying `checks` directory is self-contained; none imports production code, the independent geometric verifier, another checker, or task records. The first two have an optional numerical mode; use `-exact-only` for the theorem gates. Run with ordinary Python assertions enabled (do not use `-O`).

7.1 Prefix Gates

The file `upper_lambda.py` uses interval endpoints on the grid 10^{-50}. Each arithmetic operation rounds outwards, division across zero is rejected, and $\sqrt{n/10^{50}}$ is enclosed using the integer square root of $n10^{50}$. Two half-angle reductions express $\arctan y = 4\arctan q$ with $|q| < 1$. The first 64 alternating terms are widened by $|q|^{129}/129$ before multiplying by four. For $\log y$, take two square roots, set $q = (y^{1/4}-1)/(y^{1/4}+1)$, and use the 64-term odd series for $\log y = 8\sum_{j\ge 0} q^{2j+1}/(2j+1)$, with the corresponding tail bounded by $8|q|^{129}/[129(1-|q|^2)]$. Every operation in those formulas is itself enclosed.

At rational $x > \tau$, 100 dyadic bisections of $[0, x]$ isolate the unsquared equation $v(z, x) = 1$. An unseparated sign aborts the check. The exact primitives

$$\mathcal{F}(t,c) = \frac{(2t+c)\sqrt{t(t+c)} - c^2\log((\sqrt{t}+\sqrt{t+c})/\sqrt{c})}{4},$$

$$\mathcal{J}(v,M) = \frac{v\sqrt{M^2-v^2} + M^2\arctan(v/\sqrt{M^2-v^2})}{2}, \qquad \mathcal{H}(c,t) = c\arctan\sqrt{t/(c-t)} - \sqrt{t(c-t)}$$

have derivatives $\sqrt{t(t+c)}$, $\sqrt{M^2-v^2}$ and $\sqrt{t/(c-t)}$, respectively. They evaluate the split integrals in (10) and (12). Explicitly the block integral is

$$\begin{aligned}&\mathcal{J}(z-x/2, 1+x/2) - \mathcal{J}(-x/2, 1+x/2)\\ &\quad + \mathcal{F}(x,1) - \mathcal{F}(z,1)\\ &\quad + \mathcal{J}(x-(1+x)/2, (1+x)/2) - \mathcal{J}(z-(1+x)/2, (1+x)/2).\end{aligned}$$

The last four terms are omitted in the chord regime. Subtract $x + x^2/2$ for $x \le 1/3$, or $7/18 + 2[\mathcal{F}(x,1) - \mathcal{F}(1/3,1)]$ otherwise. For (12), $I_1 = \mathcal{H}(2+x, 1+z) - \mathcal{H}(2+x, 1)$ and $I_2 = \mathcal{H}(1+x, x) - \mathcal{H}(1+x, z)$. This gives the strict rational gates

Quantity	Lower Bound	Upper Bound
$\Phi(719/2500)$	$-1/20000$	$-1/25000$
$\Phi(2877/10000)$	$1/10000$	$11/100000$
$\Phi(4/5)$	$-1/500$	$-1/1000$
$E(1)$	$3/250$	$13/1000$

Direct directed square roots also show $719/2500 > \tau$. These four gates establish constant signs, not the continuum derivative inequalities proved in Section 4.

7.2 Shift Gates

The file `upper_alpha_minimum.py` uses exact rational arithmetic and directed roots on the grid 10^{-24}, checking each endpoint by squaring. It isolates $z(x_0)$ with 64 sign-separated bisections. On $[0, z_l]$ it integrates c and on $[z_h, x_0]$ it integrates k, using 2048 panels each. Both integrands are concave: their second derivatives are $-M^2/c^3$ for the chord and the sum of $-1/[4(u(1+u))^{3/2}]$ and $-(1+x_0)^2/[4(u(1+x_0-u))^{3/2}]$ for the chain. Thus composite trapezoids are lower bounds and midpoints are upper bounds. The omitted interval contributes between zero and $4(z_h - z_l)$. Subtract the exact diagonal and apply (14). For each D integral in (13), use 256 panels and the concavity

$$\frac{\mathrm{d}^2}{\mathrm{d}t^2}\sqrt{\frac{t}{t+c}} = -\frac{c(4t+c)}{4t^{3/2}(t+c)^{5/2}} < 0.$$

The negative wrap term is bounded in the reversed direction. The resulting outward intervals have common denominator 10^{12}:

Quantity	Lower Numerator	Upper Numerator
$E(x_0)$	-844268665	-844268070
e	-844272415	-844268070
$D(1093/10000)$	935124205	935201790
$F'(1093/10000)$	-1427184	-1344781
$D(10931/100000)$	938842182	938919761
$F'(10931/100000)$	2282349	2364747

No evaluation at an uncertified irrational root or floating quadrature is used. The analytic convexity proof supplies uniqueness.

7.3 Second and Third Width Gates

Write $\alpha_L = 1093/10000$, $\alpha_H = 10931/100000$, $x_L = 719/2500$, $x_H = 2877/10000$, and $A_L = 1+\alpha_L$, $A_H = 1 + \alpha_H$. For the second width, put $f(x, q) = \sqrt{(x + q)/(1 + x + q)} + \sqrt{(x + q)/(1 + x)}$. It increases in q and, for $q < 1$, in x; the squared summands' x derivatives are $1/(1 + x + q)^2$ and $(1 - q)/(1 + x)^2$. Consequently

$$W(43/1000) < f(x_H, (43/1000)/A_L) < 1, \quad W(87/2000) > f(x_L, (87/2000)/A_H) > 1.$$

For positive rational squared summands p, q with residual $r = 1 - p - q > 0$,

$$\operatorname{sign}(\sqrt{p} + \sqrt{q} - 1) = \operatorname{sign}(4pq - r^2). \tag{21}$$

The file `upper_boundary.py` checks the positive residuals and the respective exact signed differences

$$-\frac{3425314135183440921448074284 77}{4417899626707306405261713617638 41}, \qquad \frac{5197439453049111112711 5563}{2534384326810619083723453185 21}.$$

Also $B > 1$ and $7/160 - 87/2000 - (87/2000)^2/8 = 431/32000000 > 0$, which proves the sharpened upper bound for ϵ_b.

Put $e_L = 43/1000$, $e_U = 7/160$ and

$$V_L = A_L x_L + e_L = 9050867/25000000, \qquad V_U = A_H x_H + e_U = 362898487/10^9.$$

Thus $V_L < v < V_U$. For $W(A, v, d) = \sqrt{(v + d)/(A + v + d)} + \sqrt{(v + d)/(A + v)}$, both squared summands decrease in A and increase in v when $d < A$. The file `upper_mixed_width.py` checks

$$W(A_L, V_U, 29/5000) < 1, \qquad W(A_H, V_L, 67/10000) > 1.$$

Their signed differences (21), with positive residuals, are respectively

$$-\frac{9637455174046023891886794090683 17}{4734571866017504812540302482915910561}, \qquad \frac{3476335849729851070633990 7}{2052681163628866847433889039 69}.$$

Finally

$$\frac{27}{4000} - \frac{67}{10000} - \frac{(67/10000)^2}{8} = \frac{35511}{800000000} > 0, \qquad \frac{A_L}{3} - V_U - \frac{27}{4000} = \frac{354539}{3000000000} > 0.$$

These show that all corner evaluations are admissible and prove (20). The same margins give $A/4 < v < A/3$ as required when Lemma 6.1 is applied a second time.

8 The Fourth Block and the Final Upper Endpoint

Fix all four implicit parameters above and write $B_4 = A + w$, $\eta_0 = 1/20000$, $M = B_4 + \eta/2$. The independent rational enclosing corner $W_U = V_U + 27/4000 = 369648487/10^9$ gives, for $0 < \eta \le \eta_0$,

$$\begin{aligned} A - 3w &> 354539/10^9, & A - 3w - 4\eta &> 154539/10^9, \\ A/3 - w - \eta &> 204539/(3 \cdot 10^9), & 1 - \widehat{\alpha} - w - \eta &> 520991513/10^9, \\ 3/2 - M &> 21016513/10^9. \end{aligned} \tag{22}$$

Thus the fourth slab lies strictly before the diagonal cutoff and wrap. Only this slab changes. For every continuous, possibly nonsymmetric F, its measure increment is

$$\frac{1}{2}\int_w^{w+\eta} [F(t, A+t, A+2w+\eta-t) + F(t, A+2w+\eta-t, A+t) - 2F(t, A+t, A+t)]\,\mathrm{d}t.$$

This preserves positivity and each marginal. The shared endpoint has zero mass; the third and fourth reflections remain separate. No stationarity of earlier blocks is used to cancel their unchanged full costs.

At $t = w + s$, $X = B_4 + s$, $Y = B_4 + \eta - s$, $0 \le s \le \eta$, (22) implies

$$\frac{\sqrt{t}(\sqrt{X} + \sqrt{Y})}{\sqrt{XY}} \le 2\sqrt{(w+\eta)/B_4} < 1, \qquad X - 4t = A - 3w - 3s > 0.$$

Both the inserted and removed full maxima are strictly chord throughout the closed slab. The unnormalized increment is therefore

$$\begin{aligned} 4\pi[C_4(\eta) - C_3(\Delta_*)] &= \int_0^\eta [\sqrt{(B_4+s)(B_4+\eta-s)} - (B_4+s)]\,\mathrm{d}s \\ &= -\int_{-\eta/2}^{\eta/2} \frac{z^2}{M + \sqrt{M^2 - z^2}}\,\mathrm{d}z. \end{aligned} \tag{23}$$

The radicand is at least $B_4(B_4 + \eta) > 0$. Equivalently this is

$$(\eta/2)\sqrt{B_4(B_4+\eta)} + M^2 \arcsin(\eta/(2B_4+\eta)) - \eta M.$$

Since

$$2B_4 \le M + \sqrt{M^2 - z^2} \le 2M, \qquad \int_{-\eta/2}^{\eta/2} z^2\,\mathrm{d}z = \eta^3/12,$$

we obtain, with $J_4 = C_4 - C_3(\Delta_*)$,

$$-\frac{\eta^3}{96\pi B_4} \le J_4(\eta) \le -\frac{\eta^3}{96\pi M} < -\frac{\eta^3}{576} < 0,$$

using $M < 3/2$ and $\pi < 4$. Furthermore

$$J_4(\eta) = -\frac{\eta^3}{96\pi B_4} + \mathcal{R}_4(\eta), \qquad 0 \le \mathcal{R}_4(\eta) \le \frac{\eta^4}{192\pi B_4^2}.$$

Indeed the remainder is at most $\eta^3(1/B_4 - 1/M)/(96\pi) \le \eta^4/(192\pi B_4^2)$. This is a signed finite-interval remainder and proves the first nonzero unilateral term is cubic. It is not a negative first variation. At η_0, the strict saving is $\eta_0^3/576 = 1/4608000000000000$.

The four lengths sum to $T = w+\eta_0 < 1-\widehat{\alpha}$, so Theorem 3.1 applies with $k = 4$. Its radius error is $[1268/m+16384/(3m^2)]/(4\pi)$. The actual full roots and the odd placements obtained by deletion prove (3). This establishes the upper construction independently of any finite-size optimization. It does not establish optimality of four blocks, of the marginal relaxation, or of the global geometric problem.

8.1 Independent Finite Check of the Fourth Saving

The file `upper_fourth_block.py` verifies (22) with fractions and uses a 16-panel midpoint upper enclosure of the raw inserted full maximum, subtracting the exact diagonal integral. The branch gates above justify concavity in s. For the chord difference its derivative in B_4 is

$$\frac{2B_4+\eta}{2\sqrt{(B_4+s)(B_4+\eta-s)}} - 1 \geq 0,$$

since the numerator's square minus four times the radicand is $(\eta-2s)^2$. Therefore the upper enclosing B_4 corner gives an upper cost bound. Directed roots on a 10^{-40} grid make this an exact rational integral enclosure, independent of the antiderivative and the rationalized form in (23). The code also checks 16 corners, 448 polynomial reflection moments and 48 branch probes, as bounded bookkeeping diagnostics. It rejects zero, negative, and cutoff widths for the strict chord-slab gate. Those probes are not the proofs of arbitrary-test marginal preservation, universal recovery, or the cubic coefficient; the preceding analytic arguments supply those quantifiers.

9 Scope, Provenance, and Reproduction

This text repackages existing exact theorems and proved corollaries from the seven research sources indexed in `UPPER_DEPENDENCIES.md`. That document maps each section and checker to its original source at the frozen scientific snapshot. The two width arguments are written once with an explicit generic start, and checked under the hypotheses of both applications. No new mathematical coefficient, finite certificate, or claim of external mathematical acceptance is introduced.

From the directory containing this source, run

```
python -I -S checks/upper_lambda.py --exact-only
python -I -S checks/upper_alpha_minimum.py --exact-only
python -I -S checks/upper_boundary.py
python -I -S checks/upper_mixed_width.py
python -I -S checks/upper_fourth_block.py
```

The first four copied scripts use assertions, which must remain enabled. Their additional historical comparison outputs are not proof premises of this supplement. Optional floating diagnostics are numerical observations only. None of these checks certifies a finite Ringmin global optimum. The full all-pairs and infinite-parameter statements are proved in the text, including both cyclic arcs, actual-label recovery, all seams, the root scale before asymptotic substitution, and both parities.